

Kaggle Data Science and Machine Learning Survey

A Five Year Look at the Data Science Community, 2017 to 2021

Prepared as a summary of the Kaggle Data Science and Machine Learning Survey, covering responses collected each year from 2017 through 2021.

Overview

Every year, Kaggle asks people working in and around data science to fill out a short survey about who they are, what they know, and what tools they use. This report looks at five years of that survey, from 2017 through 2021, and pulls out the patterns that matter most. In total, the combined dataset holds 106,301 responses across 293 questions, making it one of the largest and most consistent pictures we have of the data science field over time.

The goal of this report is to give a clear, honest snapshot of the community: who is answering the survey, what background they bring, and which languages, tools, and frameworks they actually use day to day. It is written to be shared with stakeholders, used in a presentation, or referenced during an interview, so the language is kept plain and the charts are meant to speak for themselves.

About the Data

Before any analysis, the raw survey files needed some cleanup. Each year's survey used slightly different wording for the same answer, for example "Bachelor's degree" was sometimes stored with an encoding error, and coding experience was sometimes labeled "1 to 2 years" and other times "1-2 years." These entries were standardized so that responses could be compared fairly across years. Rare or overlapping country names, job titles, and tool names were also grouped together so the totals reflect real categories rather than formatting differences.

It is worth noting that some questions, including the ones about programming languages, IDEs, and machine learning frameworks, were not asked in the 2017 survey. Where that is the case, this report focuses on the years the question was actually asked, which is 2018 through 2021.

Who Took the Survey

The number of responses grew over the five years, ranging from around 16,700 in 2017 up to roughly 26,000 in 2021, with some natural year to year variation in between. This growth reflects both Kaggle's expanding user base and rising global interest in data science as a career path.

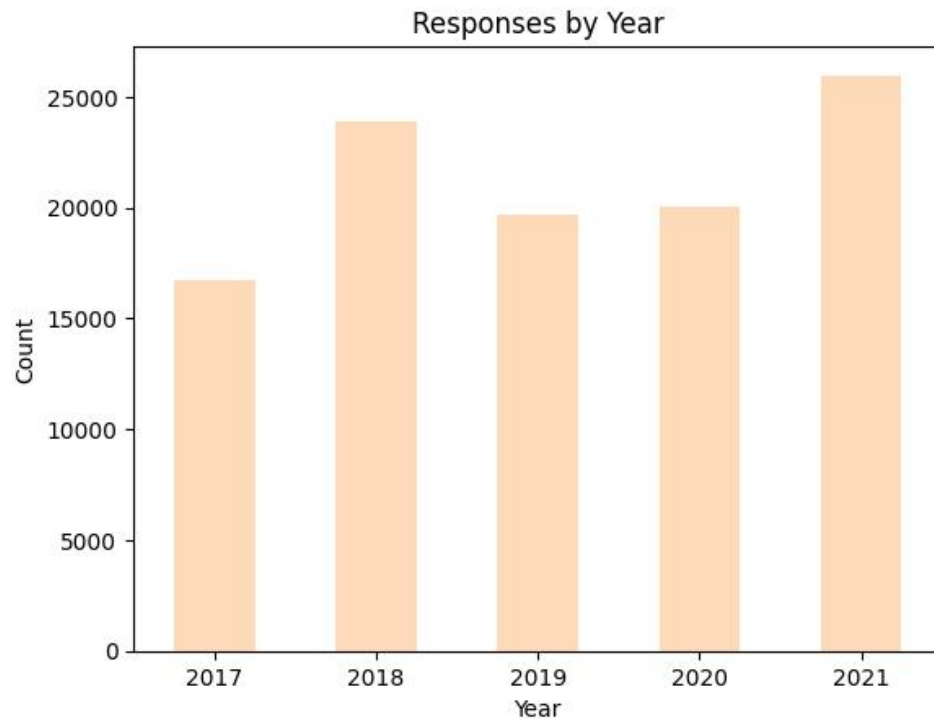


Figure 1. Number of survey responses collected each year, 2017 to 2021.

Age

The data science community skews young. Respondents aged 25 to 29 make up the single largest group at about 22 percent of all answers, and together with the 18 to 24 and 30 to 34 age bands, well over half of all respondents are under 35. Participation drops off steadily after age 40, with those 60 and older making up only a small fraction of the total.

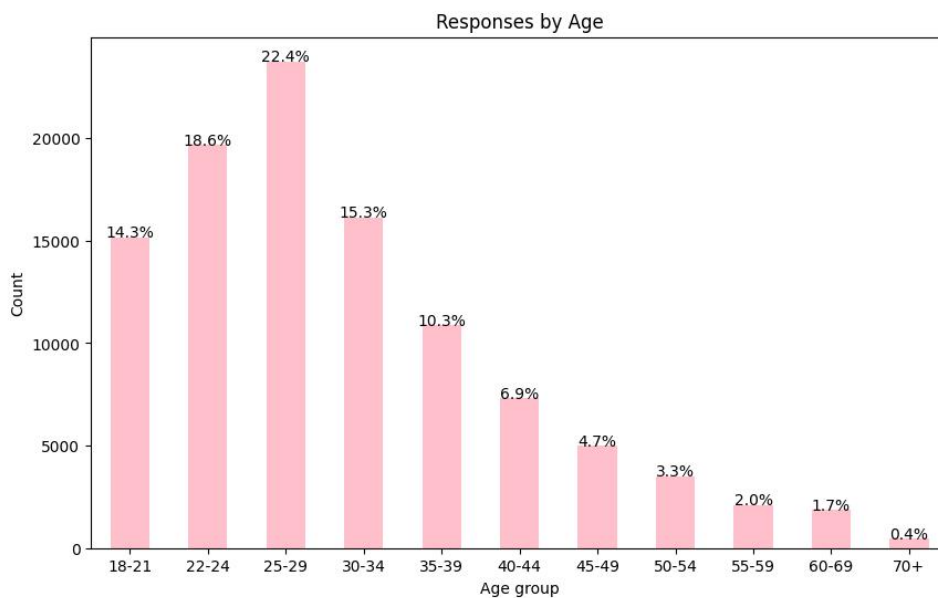


Figure 2. Distribution of respondents by age group, all years combined.

Gender

The survey continues to show a strong gender gap in the field. About 4 out of every 5 respondents identify as male, while roughly 1 in 5 identify as female, and a small remainder identify as nonbinary, prefer not to say, or self describe another way. This imbalance has held fairly steady across all five survey years, which points to a gap that has not closed much over time and is worth flagging for any organization thinking about diversity in hiring or outreach.

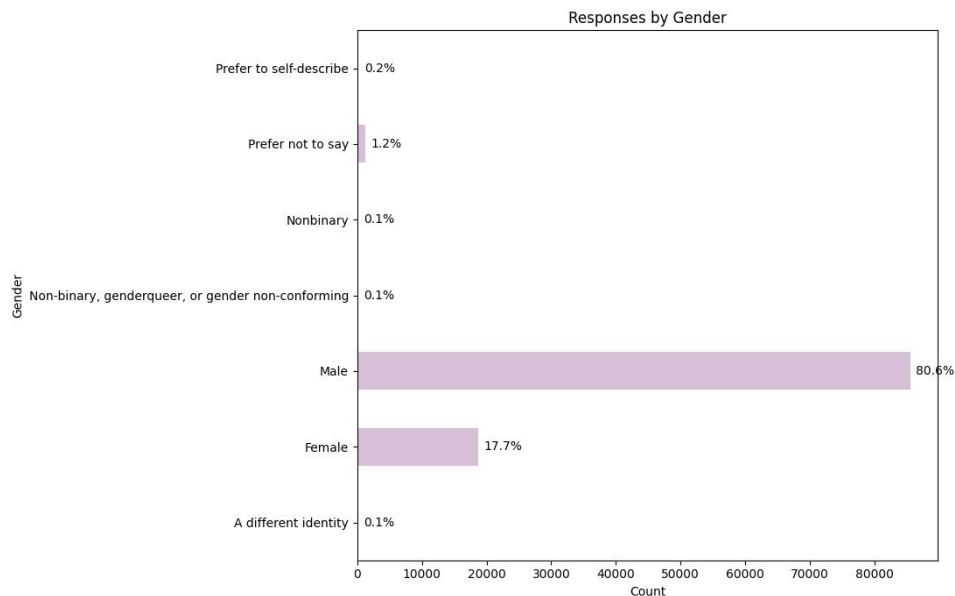


Figure 3. Distribution of respondents by gender, all years combined.

Country

Respondents come from all over the world, but two countries stand out well above the rest. India consistently sends the largest share of respondents, followed closely by the United States, and together these two countries account for well over half of all responses in most years. Brazil, China, Nigeria, Russia, and the United Kingdom round out the next tier of contributing countries, showing that the community, while global, is concentrated in a handful of large tech markets.

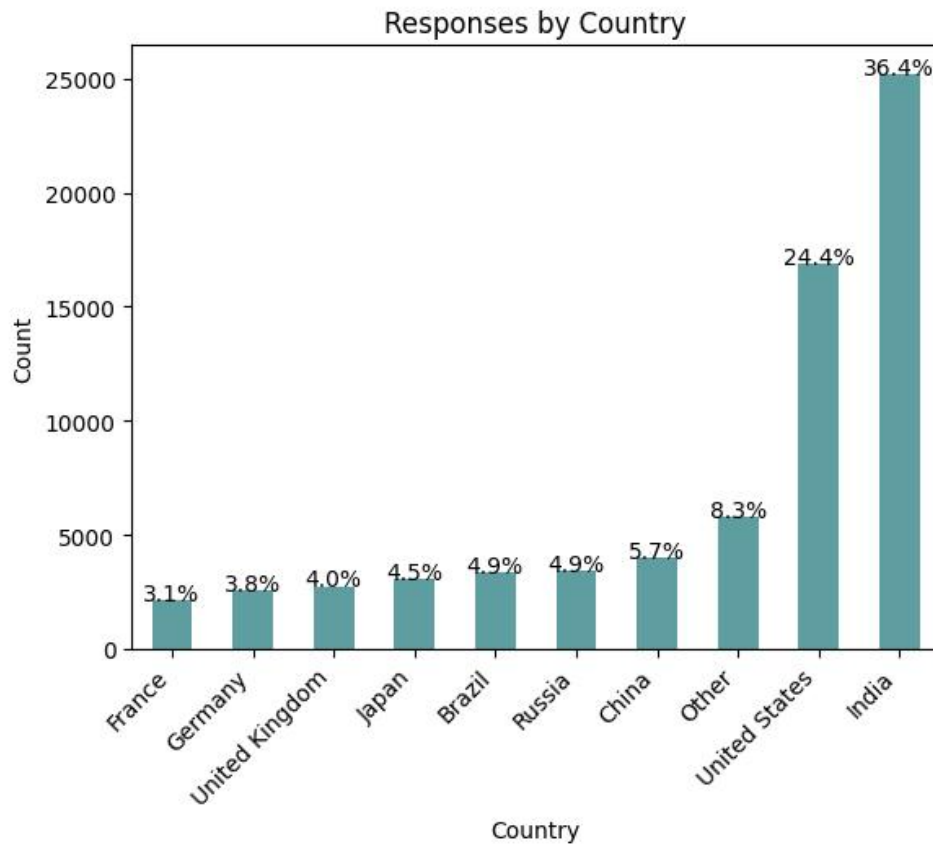


Figure 4. Top ten countries by number of respondents, all years combined.

Education

This is a highly educated group. A master's degree is the single most common qualification, held by about 43 percent of respondents, and a bachelor's degree is next at about 34 percent. Add in the 14 percent who hold a doctoral degree, and close to 9 out of 10 respondents have completed a four year degree or higher. Very few respondents report stopping their education at high school.

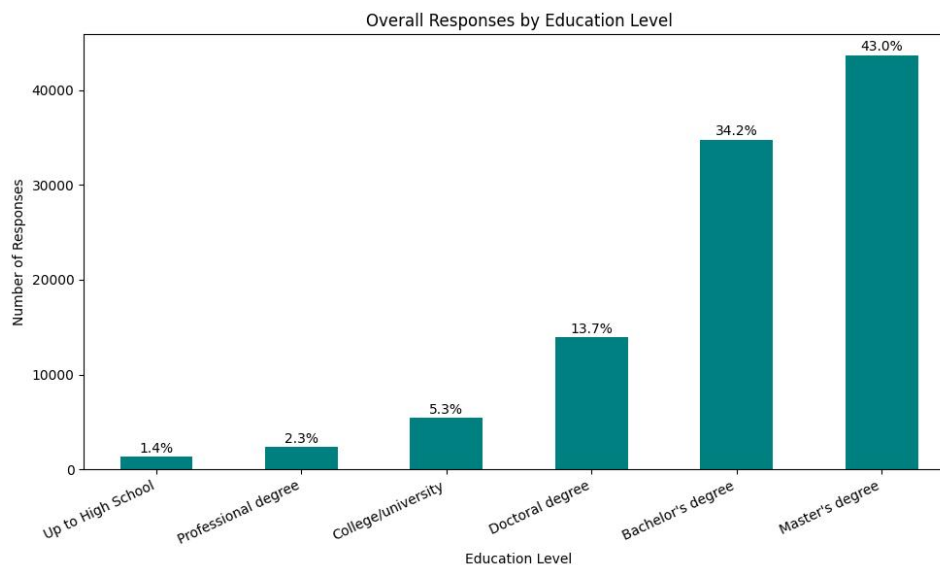


Figure 5. Highest level of education reported, all years combined.

Career Background

Current Role

Students make up the largest single group of respondents at close to 24 percent, which is an important detail to keep in mind when reading the rest of this report, since a student's tool choices and experience level can look quite different from someone working in industry. After students, Data Scientist is the most common title at about 19 percent, followed by Software Engineer, Data Analyst, and Research Scientist. Together, these five roles account for the large majority of all responses.

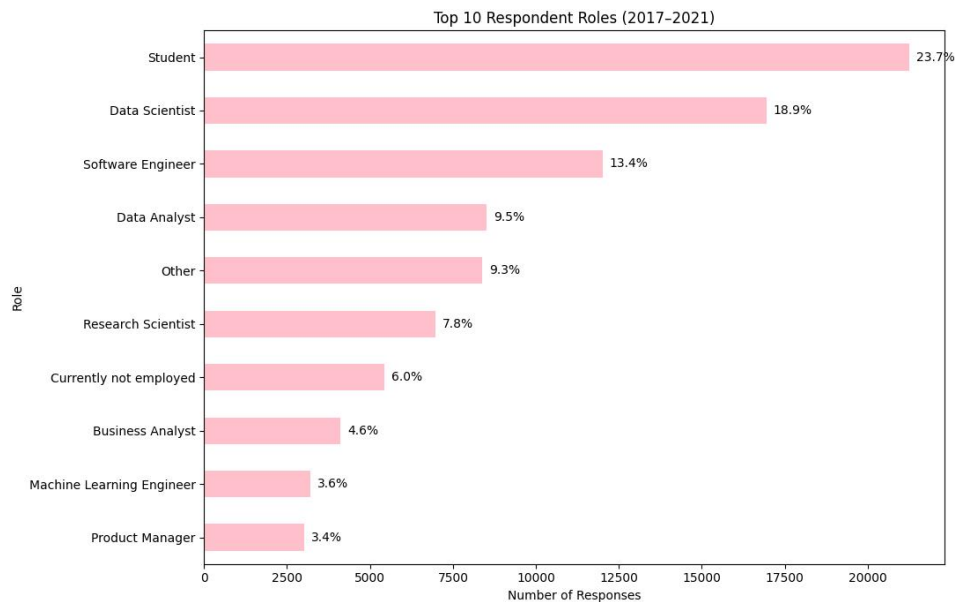


Figure 6. Top ten reported job roles, all years combined.

Coding Experience

In line with the large student population, most respondents are still early in their coding journey. Around 21 percent report less than one year of coding experience, and another 27 percent report one to two years, meaning roughly half of all respondents have two years of coding experience or less. Only about 13 percent report ten or more years of experience. This mix suggests the survey captures a field that is still growing and bringing in a steady stream of newcomers each year.

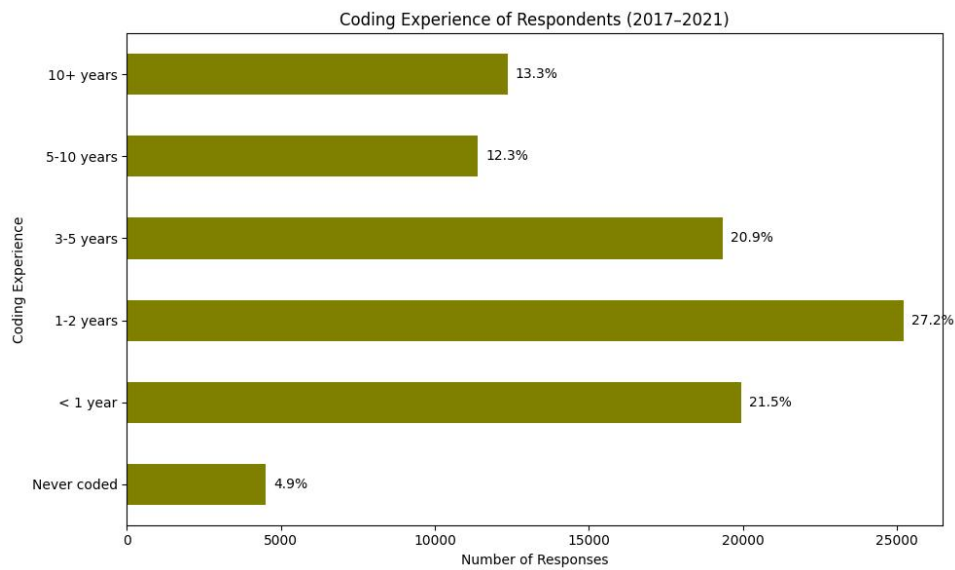


Figure 7. Coding experience of respondents, all years combined.

Tools of the Trade

Programming Languages

Python is, by a wide margin, the language of choice for this community. Across 2018 to 2021, Python was selected roughly 65,900 times, well ahead of SQL at about 33,100 and R at about 20,900. Java, JavaScript, C++, C, and MATLAB each hold a modest, fairly steady following, while newer languages like Julia and Swift remain niche choices.

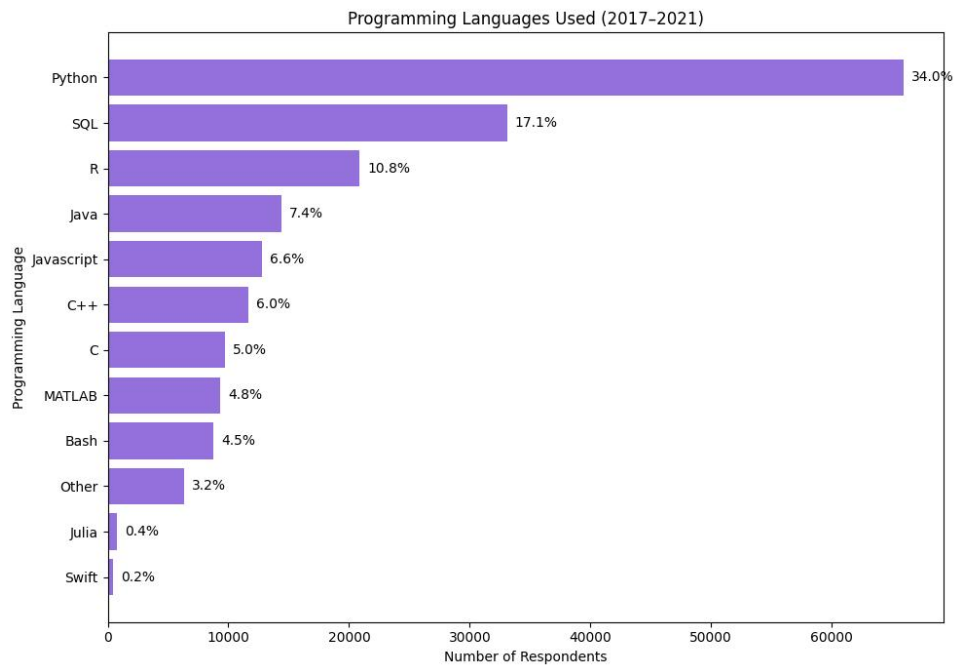


Figure 8. Programming languages used on a regular basis, 2018 to 2021 combined.

Looking at the trend year by year tells an even clearer story. Python usage grew from about 15,700 responses in 2018 to about 21,900 in 2021, a jump of close to 40 percent, and it pulled further ahead of every other language over that period. SQL also grew at a healthy pace, rising from about 8,300 to about 10,800 responses. R, on the other hand, slipped from about 6,700 responses in 2018 down to about 5,300 in 2021, suggesting a slow but steady shift in preference away from R and toward Python and SQL.

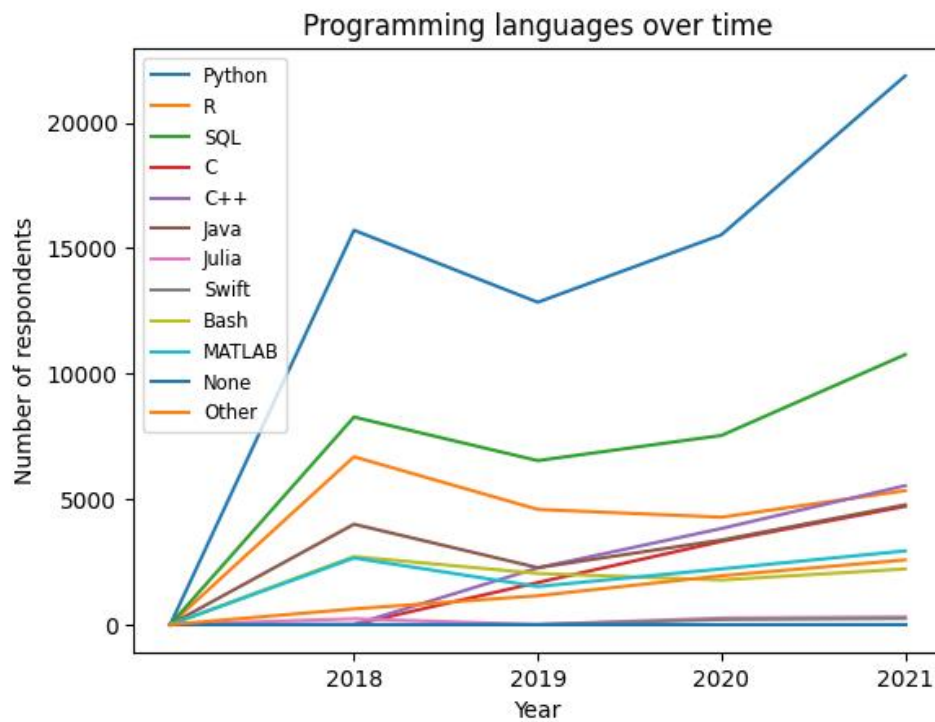


Figure 9. Programming language usage trend, 2018 to 2021.

Development Environments

Jupyter notebooks were the most used development environment overall across 2018 to 2021, with about 41,500 total mentions, ahead of Visual Studio at about 27,100 and PyCharm at about 23,900. RStudio and Notepad++ also show strong followings.

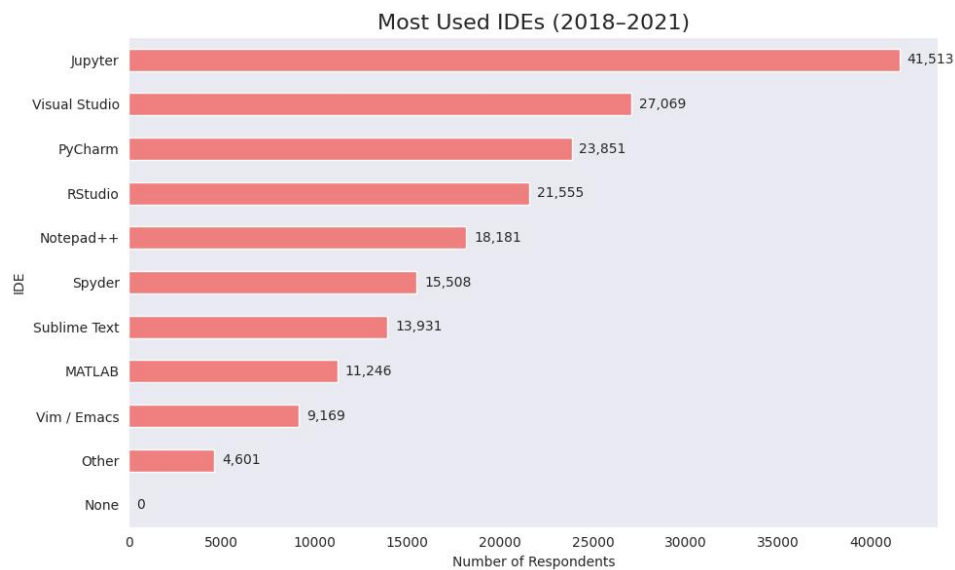


Figure 10. Most used development environments, 2018 to 2021 combined.

The year by year trend reveals a notable shift in 2021. Jupyter usage, which had led every prior year, dropped sharply from about 11,200 responses in 2020 to about 5,500 in 2021, while Visual Studio jumped from about 2,400 to about 12,300 responses over the same period, making it the most used environment in 2021 for the first time. This kind of shift is worth watching, since it may point to Visual Studio Code's growing pull as an all in one editor for both notebooks and general purpose code.

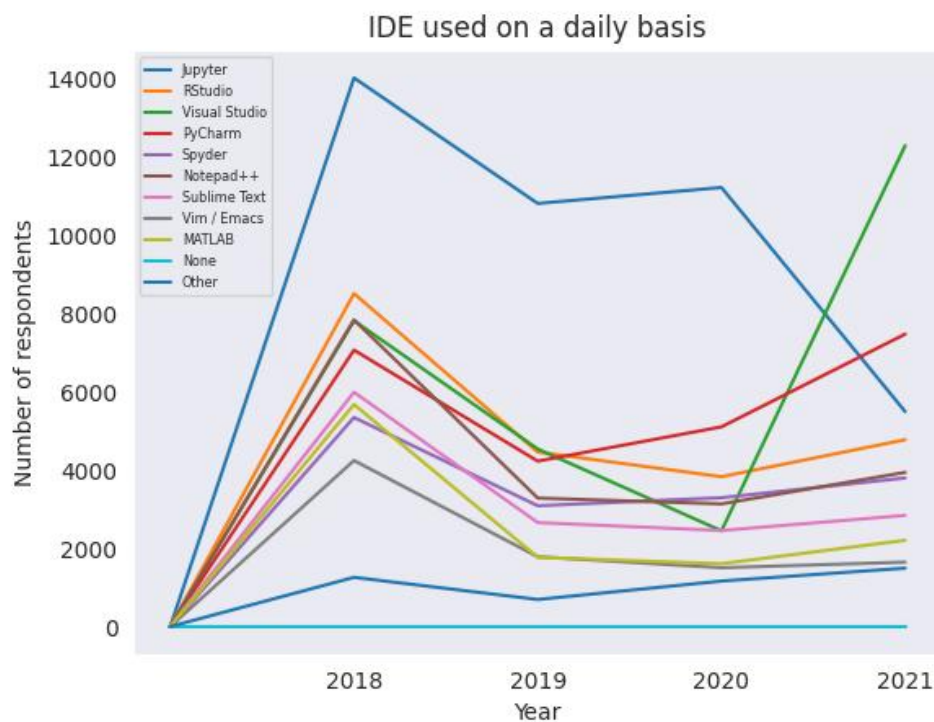


Figure 11. Development environment usage trend, 2018 to 2021.

Machine Learning and Business Intelligence Tools

Scikit-Learn is the clear favorite machine learning framework, used by about 27 percent of all framework mentions from 2018 to 2021, roughly 45,900 responses in total. TensorFlow and Keras follow next, together making up close to 35 percent of mentions, which shows that deep learning tools remain very popular even though traditional machine learning is still the most common day to day choice. Looking at growth alone, PyTorch stands out, its yearly mentions grew from about 3,800 in 2018 to about 6,100 in 2021, a rise of close to 60 percent, making it the fastest growing major framework in the survey even though it has not yet caught up to Scikit-Learn, TensorFlow, or Keras in total usage.

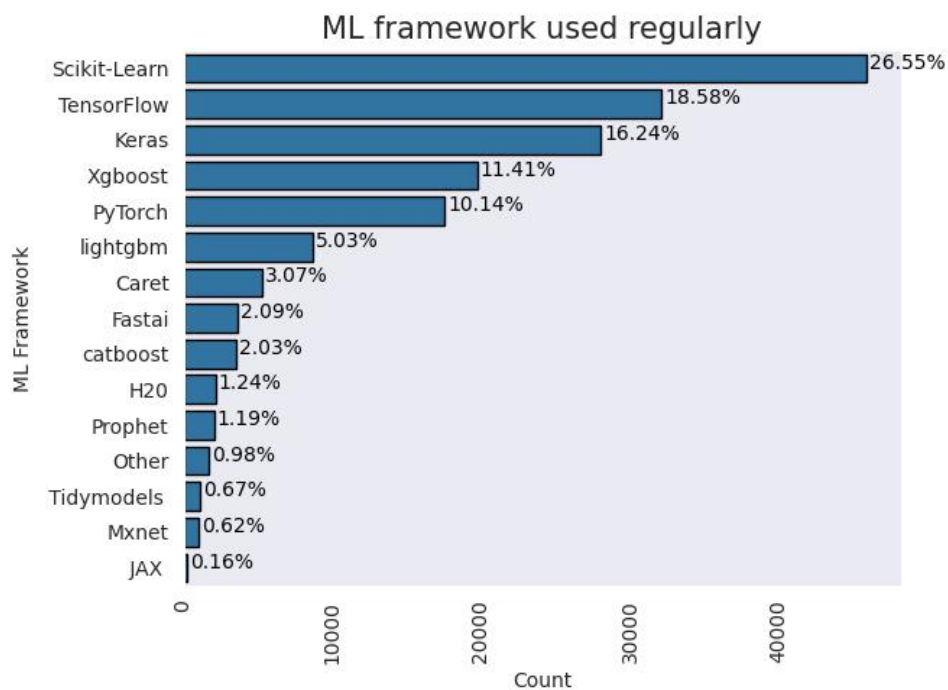


Figure 12. Machine learning frameworks used on a regular basis, 2018 to 2021 combined.

On the business intelligence side, Tableau and Microsoft Power BI dominate the field, together accounting for roughly two thirds of all business intelligence tool mentions. Google Data Studio is a distant third, and the remaining tools, including Qlik, Salesforce, and Amazon QuickSight, each hold a small single digit share. This tells a simple story: when it comes to dashboards and reporting, most of the market has settled around just two major platforms.

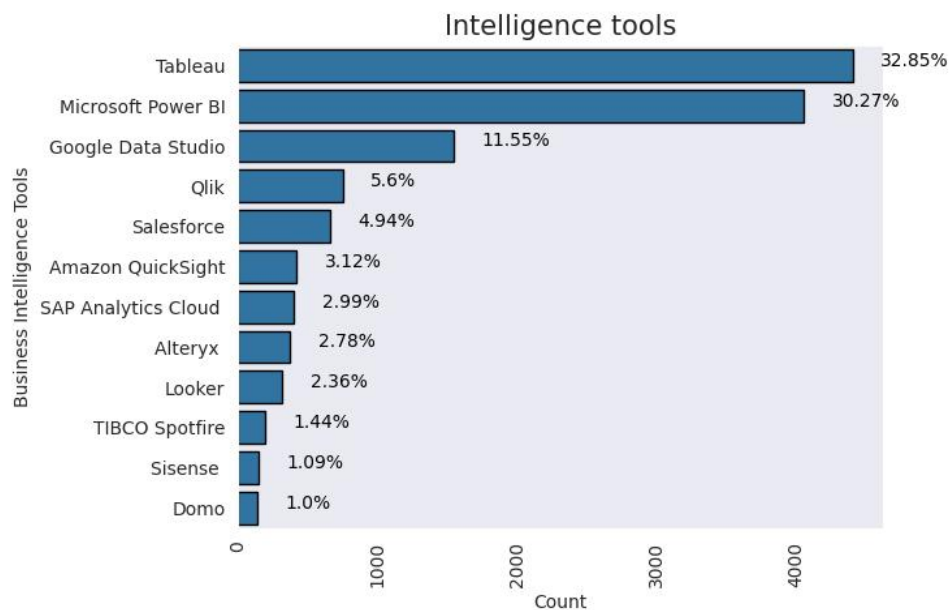


Figure 13. Business intelligence tools used on a regular basis, all years combined.

Key Takeaways

- Python and Scikit-Learn are the backbone of everyday data science work, and both have only strengthened their lead over the five year period.
- The community skews young, highly educated, and still early in its coding journey, with students forming the single largest respondent group.
- The gender gap in data science has remained wide and largely unchanged, with about four out of five respondents identifying as male every year.
- India and the United States lead in respondent volume, reflecting where much of the world's data science talent and interest is concentrated.
- PyTorch is the fastest growing machine learning framework, and Visual Studio overtook Jupyter as the leading development environment in 2021, both signs of where tool preferences may be heading next.
- Tableau and Microsoft Power BI together dominate the business intelligence space, leaving little room for smaller competitors.

Conclusion

The five years of survey data tell a consistent story about a field that is growing quickly but still settling into its habits. Despite constant talk about new deep learning tools and techniques, traditional machine learning approaches, led by Scikit-Learn, remain the everyday standard, most likely because they are easier to learn and require less infrastructure than deep learning pipelines. Python has cemented itself as the default language of the field, and the shift toward Visual Studio as a development environment suggests that tool preferences are still very much in motion.

One important caveat is that students make up a large share of respondents, and their needs, tools, and experience levels can differ quite a bit from someone working in industry. This likely pulls some results, such as coding experience and tool familiarity, toward what a learner would choose rather than what a seasoned professional would choose. Even with that in mind, the overall pattern is clear: this is a fast growing, global, and increasingly standardized field, with Python, Scikit-Learn, and Jupyter style notebooks at its center, and a steady stream of new tools working their way in each year.